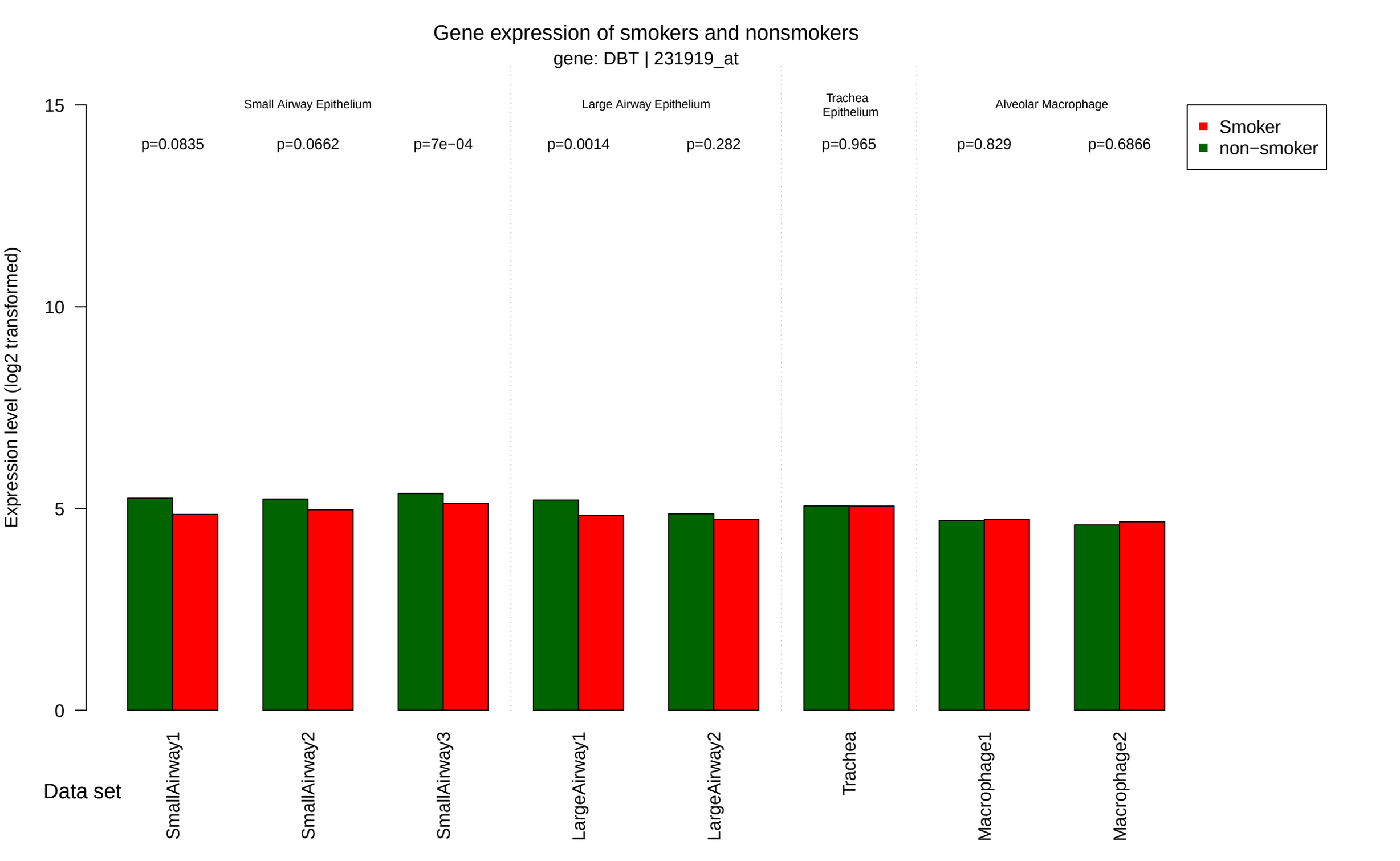
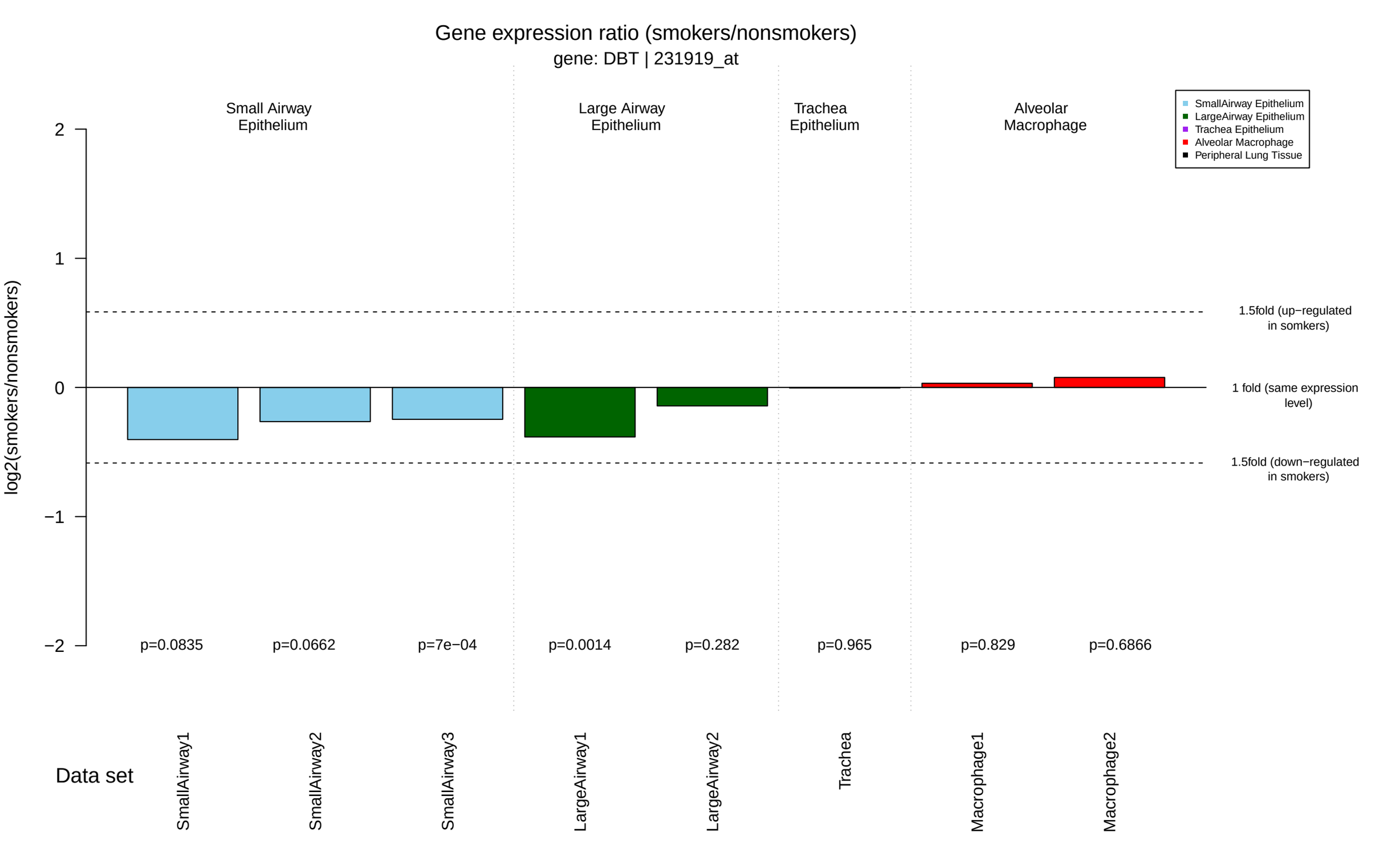
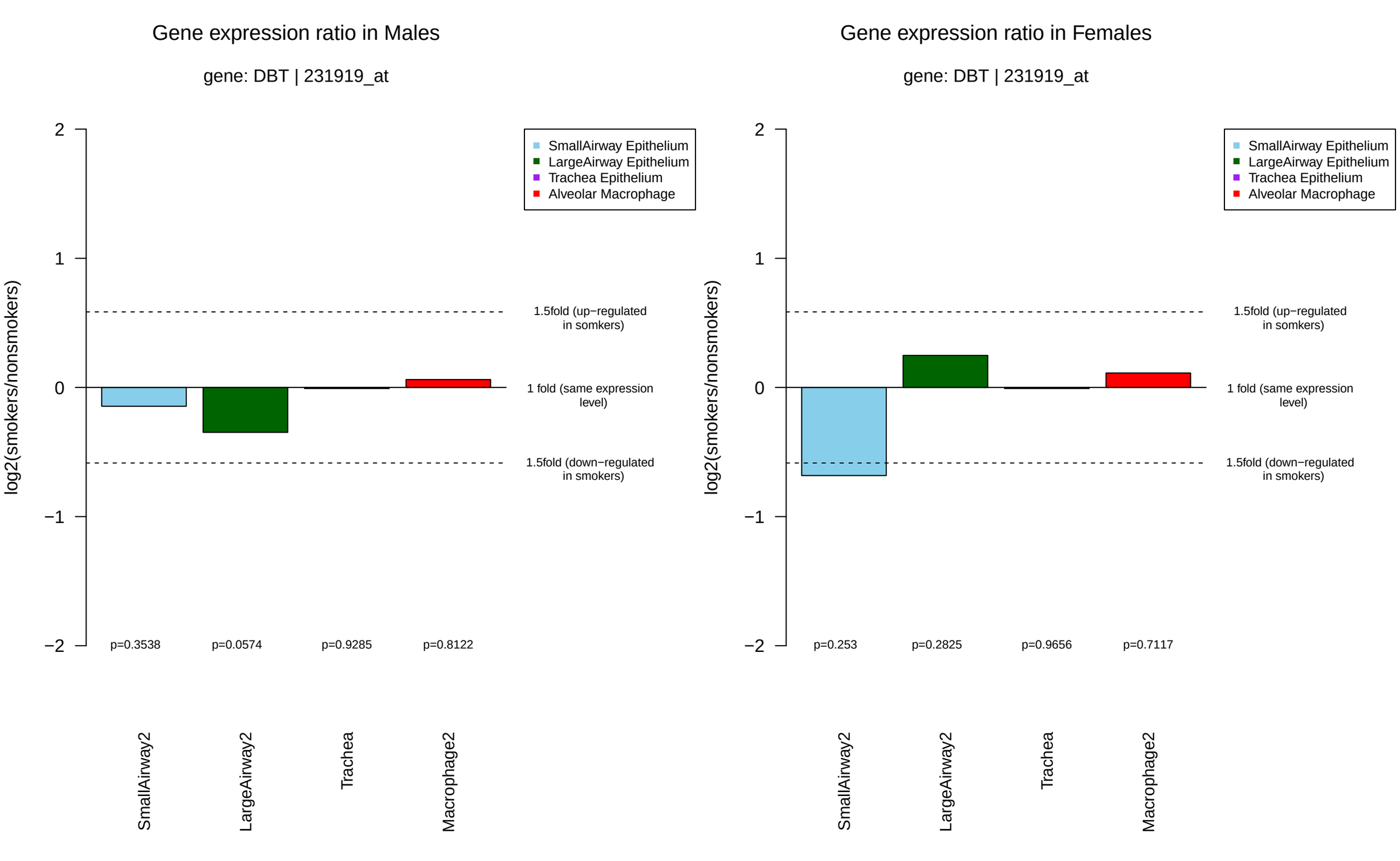
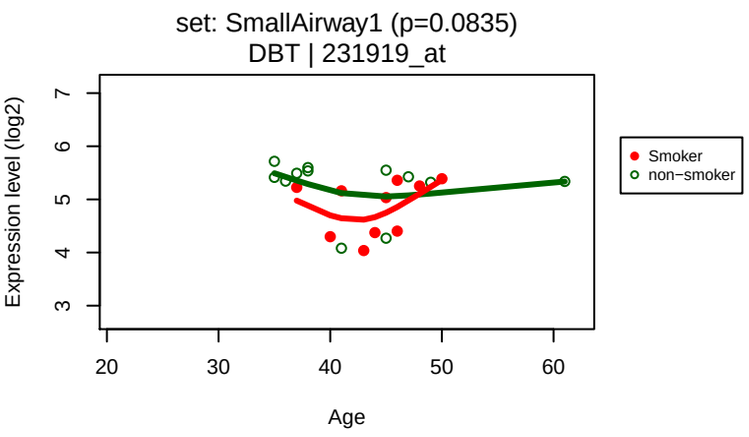
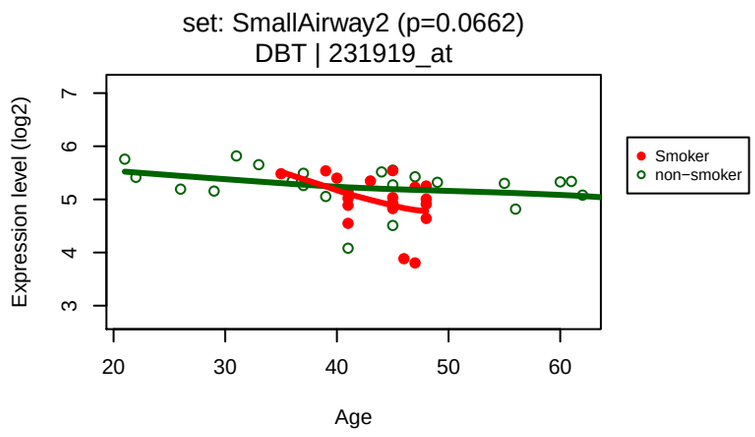
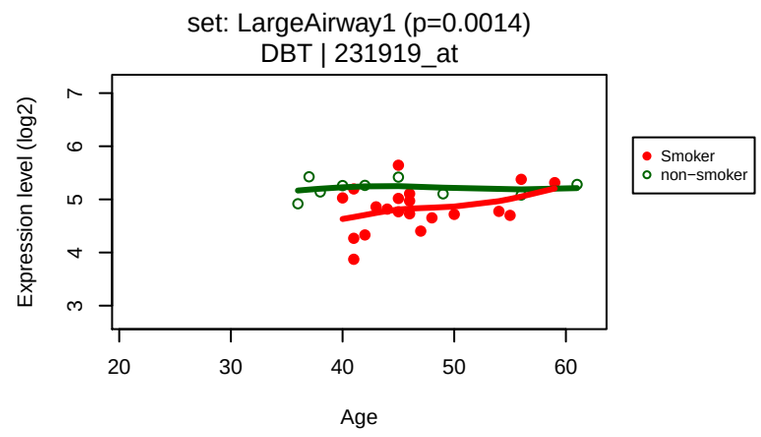

Gene expression of smokers and nonsmokers
gene: DBT | 231919_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0835
p=0.0662
p=7e−04
p=0.0014
p=0.282
p=0.965
p=0.829
p=0.6866
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

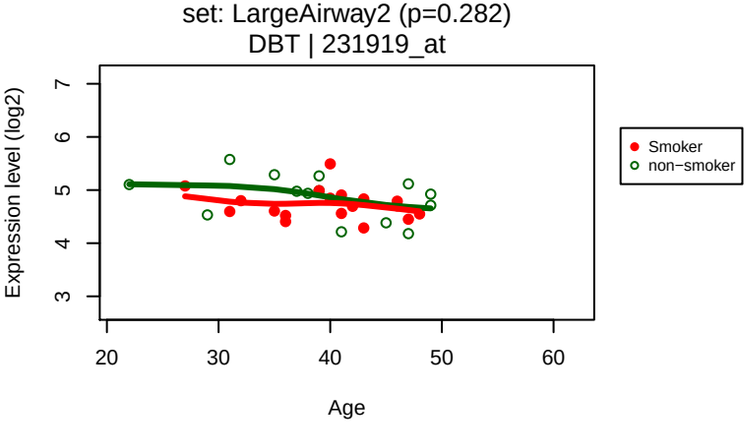

Gene expression ratio (smokers/nonsmokers)
gene: DBT | 231919_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0835
p=0.0662
p=7e−04
p=0.0014
p=0.282
p=0.965
p=0.829
p=0.6866
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

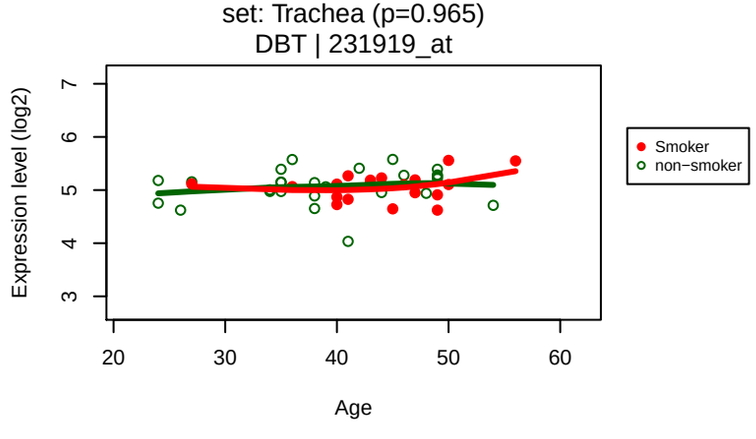

Gene expression ratio in Males
gene: DBT | 231919_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3538
p=0.0574
p=0.9285
p=0.8122
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DBT | 231919_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.253
p=0.2825
p=0.9656
p=0.7117
SmallAirway2
LargeAirway2
Trachea
Macrophage2

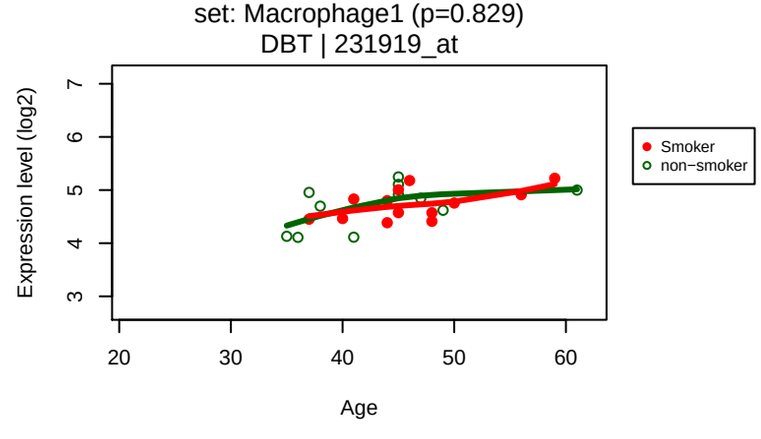

set: SmallAirway1 (p=0.0835)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker

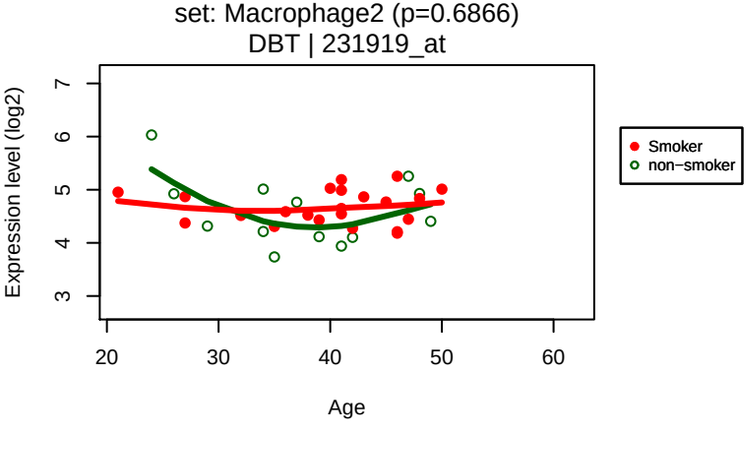

set: SmallAirway2 (p=0.0662)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker

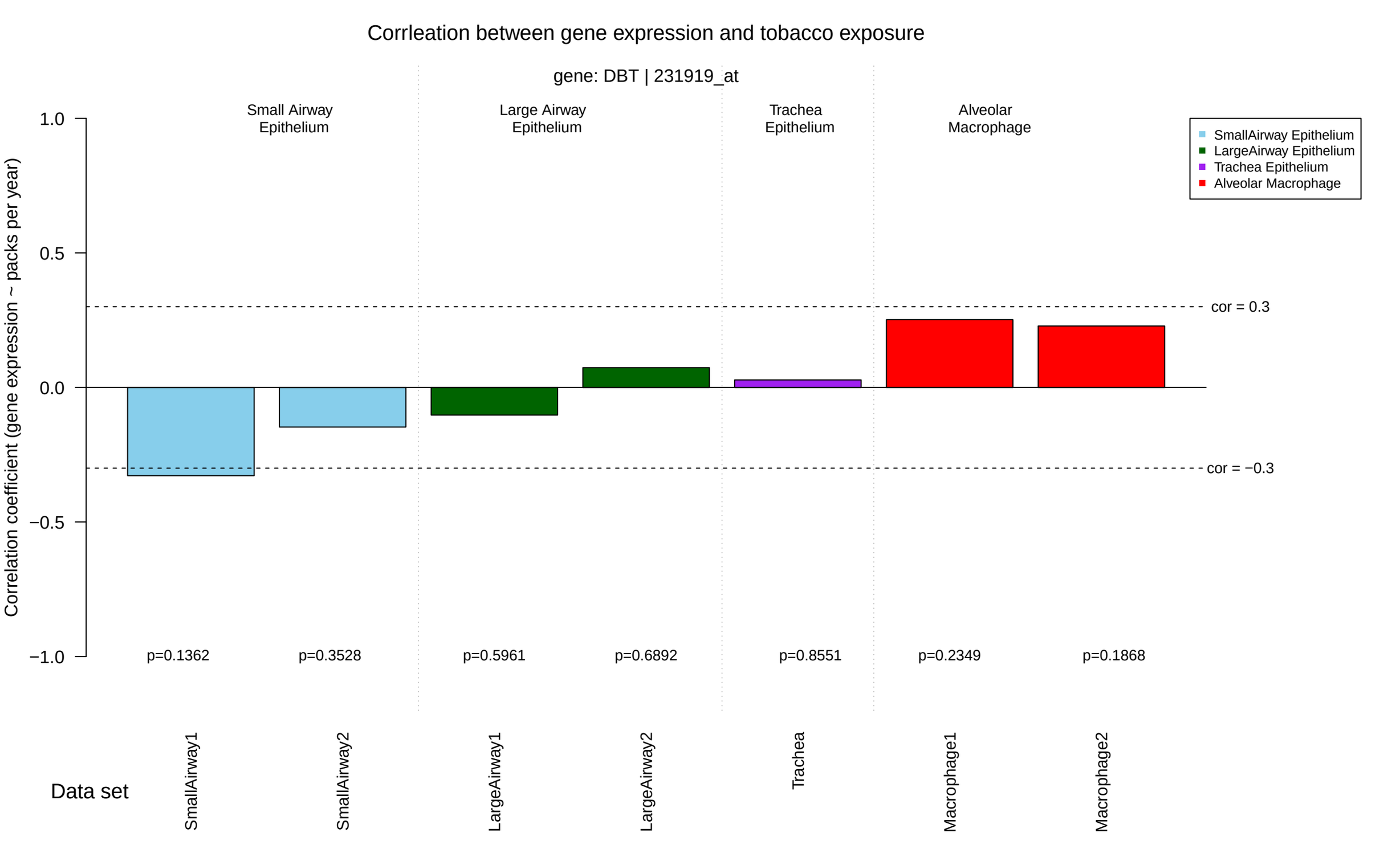

set: LargeAirway1 (p=0.0014)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.282)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.965)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.829)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6866)
DBT | 231919_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DBT | 231919_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1362
p=0.3528
p=0.5961
p=0.6892
p=0.8551
p=0.2349
p=0.1868
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2